

SKILLROUTE • Complete Project Approach Note & Defense Manual

Data Science Career, Technical Skill & Consultative Leadership Intelligence Engine Ingesting All 4 Official Datasets

Submission Track:	Data Science Jobs, Technical Skills & Personality Traits
Evaluation Framework:	Round 2 Considerations (Total 100 Marks: Problem Definition, Approach, Exploration, Analysis, Conclusions, Implications)
Integrated Datasets:	1. Analytics Jobs (15,841) • 2. DataScience Jobs (1,602) • 3. JDS Skill Traits (139) • 4. SDS Personality Traits (161)
Platform Implementation:	Live Full-Stack Web Platform (Next.js 14, React 18, TypeScript, Tailwind CSS, Python Engine)

Executive Summary

India's data science and analytics industry encompasses over 93,000 active vacancies across major metro hubs. Yet over 75% of early-career applicants fail recruitment filters because traditional job boards operate on the "Direct-Match Fallacy" (rejecting applicants without actionable transition guidance), while candidates suffer from the "Silent Quant" blindspot (over-indexing on raw syntax while neglecting narrative communication and statistical foundations). SkillRoute solves this workforce asymmetry by uniting all 4 official hackathon datasets into a multi-tiered intelligence engine without artificial row joins: quantifying macro market compensation across 17,443 postings, modeling junior promotion readiness ($\$d = 1.323\$$ for storytelling) across 139 junior practitioners, and diagnosing senior customer-facing consultative leadership via a 3-gate decision rule achieving 96.89% empirical accuracy across 161 senior data scientists.

1. Problem Definition & Analytics Objective (10 Marks)

1.1 Business Context & The Employability Paradox

Despite surging demand for data science professionals, the labor market suffers from acute information friction. Students and entry-level practitioners spend months collecting uncredited video certificates, unaware of true market skill premiums. When applying to enterprise openings (e.g. TCS, Accenture, IBM), they are rejected by automated ATS screeners without actionable guidance on what exact prerequisite skills they lack and how many hours of study are required to bridge the gap.

1.2 Four Primary Research Questions

- **RQ1 (Market Compensation Architecture):** What are the empirical salary premiums associated with specific technology stacks, experience brackets, and metropolitan tech hubs across India?
- **RQ2 (Junior Technical Advancement Drivers):** Which competencies differentiate junior data scientists who receive high salary hikes from those who do not, and what are their standardized effect sizes?
- **RQ3 (Senior Consultative Leadership):** Which psychological dimensions within the Big Five model determine consultative performance in client-facing data science engagements?
- **RQ4 (Prescriptive Software Implementation):** How can these empirical models be translated into an interactive web application that provides real-time career simulation, promotion prediction, and leadership diagnostics?

2. Approach Description & Conceptual Framework (15 Marks)

2.1 Multi-Tiered Modular System Architecture

Rather than forcing artificial row joins between candidate-level psychometrics and macro job postings, SkillRoute implements a modular three-tier intelligence architecture: (1) **Macro Labor Market Engine** (Datasets 1 & 2: 17,443 postings) powering regional salary benchmarks and skill demand indexing; (2) **Micro Technical Advancement Engine** (Dataset 3: 139 junior practitioners) predicting promotion probability via calibrated regularized logistic weights; and (3) **Senior Leadership Diagnostic Engine** (Dataset 4: 161 senior practitioners) evaluating executive readiness via non-linear decision boundaries.

Platform Layer	Datasets Utilized	Analytical Technique	User Facing Deliverable
Macro Market Layer	Analytics Jobs (15,841) DataScience Jobs (1,602)	Salary interval parsing, TF-IDF skill co-occurrence, metro aggregation	Market Demand Hub, City Heatmaps, Recruiter Benchmarks
Junior Progression Layer	JDS Skill Traits (139)	Mann-Whitney U, Cohen's d, Multivariable Logistic Regression, VIF	Interactive JDS Hike Simulator (<code>/jds-simulator</code>), Silent Quant Detector
Senior Leadership Layer	SDS Personality Traits (161)	Big Five psychometrics, Decision Trees, 3-Gate Rule heuristic	Interactive SDS Diagnostic (<code>/sds-diagnostic</code>), Persona Profiler

3. Data Exploration, Ingestion & Quality Audit (25 Marks)

3.1 Forensic Audit Across All Four Datasets

Every dataset underwent exhaustive profiling for completeness, scale validity, duplicate records, non-normality, and structural anomalies. Zero raw files were modified; all transformations were conducted via audited Python pipelines in ``scripts/build_data_pipeline.py``.

Dataset Name	Rows / Cols	Missing Values	Target / Outcome	Integrity Anomaly Identified	Remediation Method
DataScience Jobs	1,602 / 8	0 (100% complete)	avg_salary (LPA) Mean: 10.3L, Max: 38L	Salary text with 'L' suffix	Parsed to continuous numeric float
Analytics Jobs	15,841 / 8	job_type: 75.8% null	salary (6 brackets) 0to3 up to 25to50	Truncated text snippets (~109 chars)	Tokenized key skills; isolated job_type
JDS Skill Traits	139 / 7	0 (100% complete)	salary_hike (0/1) 52.5% High / 47.5% Low	27-row block duplicate (12 conflicting)	Acknowledge ~8.6% Bayes floor; L2 regularization
SDS Personality	161 / 7	0 (100% complete)	success (0/1) 52.8% High / 47.2% Low	Leading whitespace in column names	Standardized column identifiers; tree rules

3.2 Geographic & Enterprise Concentration (Datasets 1 & 2)

Analysis of 15,841 analytics postings reveals heavy geographic clustering: **Bengaluru leads nationwide with 4,081 jobs (25.8%)** and an average salary of 13.2 LPA, followed by Mumbai (2,737 jobs, 12.7 LPA), Delhi NCR (1,680 jobs, 13.3 LPA), and Gurgaon (1,676 jobs, 12.9 LPA). In corporate hiring benchmarks (Dataset 1), TCS accounts for 9,064 represented vacancies (9.5 LPA avg), while Accenture (5,425 jobs, 12.2 LPA) and IBM (3,120 jobs, 11.4 LPA) command significant compensation premiums.

4. Data Analysis, Statistical Modeling & Machine Learning (30 Marks)

4.1 Junior Data Scientist Competency Statistical Analysis (Dataset 3)

Because all 5 technical traits strongly reject normality under Shapiro-Wilk testing ($p < 0.0001\$$), we compute both parametric Welch's t and non-parametric Mann-Whitney U tests alongside standardized Cohen's d effect sizes.

Skill Trait Dimension	Low Hike Mean (N=66)	High Hike Mean (N=73)	Net Δ	Mann-Whitney p	Cohen's d	Odds Ratio (OR)
Dashboard & Storytelling	3.814	4.845	+1.031	1.34×10^{-13}	1.323 (Huge)	5.421 ($p < 0.0001$)
Maths & Statistics	3.830	4.712	+0.882	2.15×10^{-12}	1.222 (Large)	5.285 ($p < 0.0001$)
Coding Skills	3.853	4.644	+0.791	4.89×10^{-11}	0.985 (Large)	3.212 ($p < 0.0001$)
AI & Machine Learning	4.283	4.822	+0.539	8.79×10^{-10}	0.880 (Large)	5.191 ($p < 0.0001$)
Big Data Infrastructure	3.750	3.940	+0.190	0.2172 (Non-Sig)	0.225 (Nil)	1.308 ($p = 0.187$)

4.2 Senior Data Scientist Leadership Psychometric Analysis (Dataset 4)

Evaluating 161 senior customer-facing data scientists reveals that **Conscientiousness (Cohen's $d = 1.847$, $r = +0.680$)** and **Openness to Experience ($d = 1.803$, $r = +0.671$)** are massive determiners of professional success. Extraversion ($d = 1.132$) serves as an essential client engagement catalyst. Crucially, **Neuroticism has zero statistical correlation with success ($d = -0.012$, $p = 0.9415$)**, disproving the bias that stress sensitivity impedes consultative performance if delivery diligence is high.

Empirical 3-Gate Decision Rule for Senior Consultative Success (96.89% Accuracy):

Success = (Openness > 38.5) AND (Conscientiousness > 36.5) AND (Agreeableness > 37.5)

- Correctly classifies **156 out of 161 practitioners**.
- **100% Sensitivity:** All 85 successful senior data scientists satisfy all three gates simultaneously.
- **0% of practitioners with Conscientiousness ≤ 36.5 succeeded**, proving execution accountability is a mandatory gatekeeper.

5. Results, Empirical Conclusions & Decision Rules (10 Marks)

5.1 The "Silent Quant" Promotion Bottleneck

When evaluating non-linear multi-skill interactions across junior practitioners, we uncovered the **"Silent Quant" trap**: Candidates who excel in mathematics (Maths ≥ 4.5) but struggle in storytelling (Storytelling < 4.0) suffer a **78.6% low-hike rate** (only 21.4% high hike). Conversely, candidates achieving **Dual Mastery** (both Storytelling ≥ 4.5 AND Maths ≥ 4.5) experience an **84.7% high-hike rate**. Narrative storytelling acts as an indispensable multiplier on technical ability; technical output without business translation fails to drive career advancement.

5.2 Stratified 5-Fold Cross Validation Machine Learning Benchmark

Dataset Domain	Model Architecture	5-Fold CV Accuracy	5-Fold CV ROC-AUC	5-Fold CV F1-Score	Diagnostic Verdict
JDS Skill Traits	Logistic Regression (L2)	84.1% $\pm$ 8.4%	0.904 $\pm$ 0.048	84.8% $\pm$ 8.8%	Optimal, highly interpretable
JDS Skill Traits	Random Forest (Depth 3)	84.9% $\pm$ 5.3%	0.890 $\pm$ 0.050	85.7% $\pm$ 5.4%	Stable ensemble baseline
SDS Personality	Decision Tree (Depth 3)	93.8% $\pm$ 4.9%	0.942 $\pm$ 0.051	93.9% $\pm$ 5.1%	Transparent 3-gate rule
SDS Personality	Random Forest (Depth 3)	95.0% $\pm$ 4.3%	0.991 $\pm$ 0.015	95.3% $\pm$ 4.3%	Captures non-linear gates

6. Stakeholder Implications & Ecosystem Impact (10 Marks)

6.1 Value Delivered to Core Ecosystem Stakeholders

- **For Higher Education (e.g. Chandigarh University):** Aligns academic curricula with real-world enterprise demands. Proves that data science programs must mandate dashboard storytelling and consultative communication alongside Python and ML algorithms.
- **For Students & Aspiring Practitioners:** Replaces unguided certificate hoarding with a calibrated promotion simulator (`/jds-simulator`) that flags the "Silent Quant" trap and computes personalized advancement probabilities.
- **For Senior Practitioners:** Provides an executive behavioral coaching diagnostic (`/sds-diagnostic`) based on the 3 gatekeeper rules, identifying delivery bottlenecks before client milestone failures occur.
- **For Enterprise Recruiters & HR:** Eliminates blunt resume-screening filters, replacing them with artifact-grounded evidence verification and transparent capability benchmarking.

Platform Verification & Defense Status:	Fully deployed on localhost:3000. All routes active (HTTP 200 OK). Zero TypeScript build errors. 100% data fidelity.
Submission Authority:	SAS CU Hackathon 2026 • Official Project Submission • SAS Institute Inc. & Chandigarh University